

# CyberDefenders - DanaBot Malware Investigation Walkthrough

---

## Difficulty

Easy

## Platform

CyberDefenders

## Goal

Create a beginner-friendly walkthrough where another user can follow the complete investigation process and finish the challenge in around 15–25 minutes.

## 1. Challenge Overview

This challenge focuses on investigating a DanaBot malware infection using a PCAP file.

Objectives:

- Analyze network traffic
- Extract malicious files from HTTP traffic
- Analyze malware behavior
- Identify Indicators of Compromise (IOCs)
- Calculate file hashes

## 2. Tools Used

Wireshark

Kali Linux

VS Code

Linux command line tools

CyberDefenders Lab Environment

## 3. PCAP Analysis Using Wireshark

Open the PCAP file in Wireshark.

Navigate: Wireshark → File → Export Objects → HTTP.

HTTP object extraction was used to identify suspicious transferred files. A suspicious file named login.php was found and exported for further analysis.

## 4. Extracted File Analysis in Kali Linux

The exported file was analyzed in Kali Linux.

Example path:

```
/home/kali/Downloads/temp_extract_dir/temp_extract_dir/login.php
```

```
find ~ -name "login.php"
```

Open the file:

```
cat login.php
```

The content was not a normal PHP file. It contained heavily obfuscated JavaScript with random variable names, encoded strings, compressed one-line JavaScript, and hidden malware functions.

## 5. JavaScript De-obfuscation Using VS Code

The extracted JavaScript was intentionally obfuscated using names such as `_0x23c2` and `_0x57c2`. Open `login.php` in VS Code and format it using Right Click → Format Document or Shift + Alt + F. Formatting makes the code readable.

- Generate a random DLL filename.
- Create an HTTP request.
- Download a second-stage payload.
- Save the payload locally.
- Execute the DLL.
- Delete itself after execution.

```
http://soundata.top/resources.dll
```

```
rundll32.exe /B <malicious_dll>,start
```

```
DeleteFile(WScript.ScriptFullName)
```

This confirmed `login.php` was actually a malicious JavaScript downloader.

## 6. Identifying Initial Access File

The malicious initial access file was `allegato_708.js`. The exported HTTP object appeared as `login.php`, but the original malicious file name was `allegato_708.js`.

## 7. SHA-256 Hash Calculation

```
sha256sum allegato_708.js
```

Purpose: create a unique fingerprint of the malware sample.

## 8. Execution Process Identification

Execution process: rundll32.exe, used to execute the downloaded DLL payload.

```
rundll32.exe /B <malicious_dll>,start
```

## 9. Second Malicious File Identification

The downloader retrieved resources.dll from <http://soundata.top/resources.dll>. Extension: .dll

## 10. Extracting resources.dll

Use Wireshark: File → Export Objects → HTTP. Save as ~/Desktop/resources.dll.

```
ls -la
```

## 11. MD5 Hash Calculation

```
md5sum resources.dll
```

MD5: e758e07113016aca55d9eda2b0ffeebe

## 12. Challenge Answers Table

| Question                        | Answer                           |
|---------------------------------|----------------------------------|
| Attacker IP Address             | 62.173.142.148                   |
| Initial Access File             | allegato_708.js                  |
| Initial File SHA-256            | Calculated using sha256sum       |
| Execution Process               | rundll32.exe                     |
| Second Malicious File Extension | .dll                             |
| Second File MD5                 | e758e07113016aca55d9eda2b0ffeebe |

## 13. Attack Chain Diagram

Attacker IP

↓

allegato\_708.js

↓

Obfuscated JavaScript Downloader

↓

resources.dll

↓

rundll32.exe Execution

↓

DanaBot Malware Infection

## 14. Commands Reference

```
find ~ -name "login.php"
```

```
cat login.php
```

```
ls -la
```

```
sha256sum allegato_708.js
```

```
md5sum resources.dll
```

## 15. Key Takeaways

This challenge provides practical experience in PCAP analysis, Wireshark HTTP extraction, malware artifact analysis, JavaScript de-obfuscation, IOC extraction, file hashing, and the SOC analyst investigation workflow.